lecture 20 : examples
19/4/23

a) find the points of max/min of $f(x,y) = xy$ over the circle $x^2+y^2=1$

b) find the points of max/min of $f(x,y,z) = y+z$ on the sphere $x^2+y^2+z^2=1$

c) find the closest and furthest point from $(0,0)$ on the curve $2x^2+3y^2-1=0$

$f(x,y) = xy \quad g(x,y) = x^2+y^2-1=0$

we seek (x,y), λ s.t. $\begin{cases} f_x(x,y) = \lambda g_x(x,y) \\ f_y(x,y) = \lambda g_y(x,y) \\ g(x,y) = 0 \end{cases}$

so $\begin{cases} y = \lambda 2x & ① \\ x = \lambda 2y & ② \\ x^2+y^2-1=0 & ③ \end{cases}$

remember: any points you find <u>must</u> satisfy all <u>3</u> conditions

sub ① into ② : $x = 2\lambda(2\lambda x) = 4\lambda^2 x \Rightarrow x - 4\lambda^2 x = 0$

factorize: $x(1-4\lambda^2) = 0 \Rightarrow \boxed{\cancel{x=0}}$ or $1-4\lambda^2=0 \Rightarrow \lambda^2 = \frac{1}{4}$
$\Rightarrow \boxed{\lambda = \pm 1/2}$

if <u>$x=0$</u> ① ~~and~~ or ② $\Rightarrow y=0$. But does $\cancel{(0,0)}$ satisfy all?
no!

③: $0^2+0^2-1=0 \Rightarrow -1=0$ #

sub: ② into ①: $y = 2\lambda(2\lambda y) \Rightarrow \boxed{\cancel{y=0}}$ or $\boxed{\lambda = \pm 1/2}$

if <u>$y=0$</u> ① or ② $\Rightarrow x=0$ #

what do we get if we sub $\boxed{\cancel{x=0}}$ or $\boxed{\cancel{y=0}}$ into ③?

$0^2+y^2-1=0 \Rightarrow y = \pm 1$. ~~Does~~ Do $\cancel{(0,1)}$, $\cancel{(0,-1)}$ work?

$\begin{cases} 1 = 2\lambda \cdot 0 & 1 \neq 0 \\ 0 = 2\lambda \cdot 1 & \checkmark \\ 0^2+1^2-1=0 & \checkmark \end{cases}$

if $y=0$: $x^2+0^2-1=0 \Rightarrow x = \pm 1$. But $(1,0)$, $(-1,0)$?

this causes a contradiction in ②

if $\lambda = \frac{1}{2}$

$$\begin{cases} y = \lambda 2x & ① \\ x = \lambda 2y & ② \\ x^2+y^2-1=0 & ③ \end{cases}$$

$$\begin{cases} y = x \\ x = y \\ x^2+y^2-1=0 \end{cases}$$

use ③: $x^2+x^2-1=0 \Rightarrow 2x^2-1=0 \Rightarrow x^2=\frac{1}{2} \Rightarrow x = \pm\frac{1}{\sqrt{2}}$

since $y=x$, $y=\pm\frac{1}{\sqrt{2}}$

$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, $\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ satisfy all 3 conditions

can write as $\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right)$

if $\lambda = \frac{-1}{2}$
$$\begin{cases} y = -x \\ x = -y \\ x^2+y^2-1=0 \end{cases}$$

③ $\Rightarrow x^2+(-x)^2-1=0 \Rightarrow x^2+x^2-1=0 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$

$y=-x$ so $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ and $\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ satisfy all 3 conditions

you can write these as $\left(\pm\frac{1}{\sqrt{2}}, \mp\frac{1}{\sqrt{2}}\right)$

which points lead to min/max of $f(x,y) = xy$
evaluate:

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$$

$$f\left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) = \frac{1}{2}$$

$$f\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) = \frac{-1}{2}$$

$$f\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{-1}{2}$$

$\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$ are points of maximum

$\left(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{\sqrt{2}}\right)$ are points of minimum

$f(x,y,z) = y+z \qquad h(x,y,z) = x^2+y^2+z^2-1 = 0$

we seek (x,y,z), λ s.t. $\begin{cases} f_x = \lambda h_x \\ f_y = \lambda h_y \\ f_z = \lambda h_z \\ h = 0 \end{cases}$

so $\begin{cases} 0 = \lambda 2x & ① \Rightarrow \cancel{\lambda = 0} \text{ or } \boxed{x=0} \text{ always} \\ 1 = \lambda 2y & ② \Rightarrow y = \frac{1}{2\lambda} \\ 1 = \lambda 2z & ③ \Rightarrow z = \frac{1}{2\lambda} \\ x^2+y^2+z^2-1=0 & ④ \end{cases}$ $\Big\} \Rightarrow z = y$

$④ \Rightarrow 0^2 + y^2 + y^2 - 1 = 0 \Rightarrow 2y^2 - 1 = 0 \Rightarrow y = \pm\frac{1}{\sqrt{2}} = z$

$\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ satisfy all 4 conditions

in this case, $\lambda = \frac{1}{2y} = \frac{1}{2z} = \frac{1}{2(\pm 1/\sqrt{2})} = \frac{1}{\pm\sqrt{2}} = \pm\frac{1}{\sqrt{2}}$

$④ \Rightarrow 0^2 + \left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 - 1 = 0 \Rightarrow \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} - 1 = 0$

$\Rightarrow \frac{2}{4\lambda^2} - 1 = 0 \Rightarrow \frac{1}{2\lambda^2} = 1 \Rightarrow 1 = 2\lambda^2 \Rightarrow \lambda^2 = \frac{1}{2} \Rightarrow \lambda = \pm\frac{1}{\sqrt{2}}$

$y = \frac{1}{2\lambda} = \pm\frac{1}{\sqrt{2}}$, $z = \pm\frac{1}{\sqrt{2}}$

$$0 = \lambda 2x \quad (1)$$
$$1 = \lambda 2y \quad (2)$$
$$1 = \lambda 2z \quad (3)$$
$$x^2+y^2+z^2-1=0 \quad (4)$$

(3) $\Rightarrow \quad \lambda = \frac{1}{2z}$

sub into (2): $1 = \frac{1}{\cancel{2}z} \cancel{2}y \Rightarrow 1 = yz$

$y = \frac{1}{z} \quad (z \neq 0)$

(4): $0^2 + \left(\frac{1}{z}\right)^2 + z^2 - 1 = 0$ solve for z?

$\frac{1}{z^2} + z^2 - 1 = 0 \quad \times z^2 \Rightarrow 1 + z^4 - z^2 = 0$

$\Rightarrow z^4 - z^2 + 1 = 0 \Rightarrow (z^2)^2 - (z^2) + 1 = 0$

$z^2 = \frac{1 \pm \sqrt{1 - 4 \cdot 1 \cdot 1}}{2 \cdot 1} = \frac{1 \pm \sqrt{-3}}{2}$ not real!

$f = y + z$

$f\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$.

$f\left(0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right) = \frac{-1}{\sqrt{2}} + \frac{-1}{\sqrt{2}} = -\frac{2}{\sqrt{2}} = -\sqrt{2}$

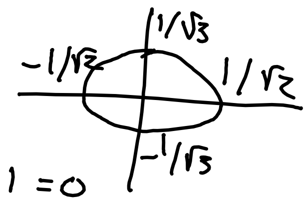
$\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ is the point of maximum

$\left(0, \frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ is the point of minimum find the points nearest and furthest from $(0,0)$

on the curve $2x^2+3y^2-1=0 \Rightarrow \frac{x^2}{1/2}+\frac{y^2}{1/3}=1$

what are we optimizing? distance

optimize $\sqrt{x^2+y^2}$



what is the constraint? $g(x,y)=2x^2+3y^2-1=0$

we can equivalently optimize $\boxed{x^2+y^2}$

square-root function is constantly increasing (monotonic)

$$\boxed{f(x,y)=x^2+y^2}$$

we seek $(x,y),\lambda$ s.t. $\begin{cases} f_x=\lambda g_x \\ f_y=\lambda g_y \\ g(x,y)=0 \end{cases}$

so $\begin{cases} 2x=\lambda 4x & (1) \\ 2y=\lambda 6y & (2) \\ 2x^2+3y^2-1=0 & (3) \end{cases}$

$$\begin{cases} 2x = \lambda 4x & (1) \\ 2y = \lambda 6y & (2) \\ 2x^2 + 3y^2 - 1 = 0 & (3) \end{cases}$$

(1) $\Rightarrow 2x - 4\lambda x = 0 \Rightarrow x(2 - 4\lambda) = 0$

(2) $\Rightarrow 2y - 6\lambda y = 0 \Rightarrow y(2 - 6\lambda) = 0$

(1) $\Rightarrow \boxed{x = 0}$ or $\boxed{\lambda = \frac{1}{2}}$. if $\lambda = \frac{1}{2}$. (2) $\Rightarrow 2y = \frac{1}{2} \cdot 6y = 3y$

$\Rightarrow y = 0$

(2) $\Rightarrow \boxed{y = 0}$ or $\boxed{\lambda = 1/3}$

if $\lambda = \frac{1}{3}$ (1) $\Rightarrow 2x = 4 \cdot \frac{1}{3} x \Rightarrow x = 0$

we can see that $(0,0)$ will not satisfy (3).

- if $\underline{x = 0}$. use (3): $2 \cdot 0^2 + 3y^2 - 1 = 0 \Rightarrow 3y^2 - 1 = 0$

$\Rightarrow y^2 = \frac{1}{3}$ so $y = \pm \frac{1}{\sqrt{3}}$ so $\left(0, \frac{1}{\sqrt{3}}\right), \left(0, \frac{-1}{\sqrt{3}}\right)$

satisfy? (1)✓ (3)✓✓ (2)? $2 \cdot \frac{1}{\sqrt{3}} = \lambda \cdot 6 \cdot \frac{1}{\sqrt{3}} \Rightarrow \lambda = \frac{1}{3}$ nothing new

$\left(0, \pm\frac{1}{\sqrt{3}}\right)$ satisfy all 3 conditions

- if $\underline{y = 0}$: use (3): $2x^2 + 3(0)^2 - 1 = 0 \Rightarrow 2x^2 - 1 = 0$

$\Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm\frac{1}{\sqrt{2}}$ so $\left(\frac{1}{\sqrt{2}}, 0\right), \left(\frac{-1}{\sqrt{2}}, 0\right)$

satisfy? (3)✓ (2)✓ (1)? $2\left(\frac{1}{\sqrt{2}}\right) = \lambda 4 \left(\frac{1}{\sqrt{2}}\right) \Rightarrow \lambda = \frac{1}{2}$ fine

$\left(\pm\frac{1}{\sqrt{2}}, 0\right)$ satisfy all 3 conditions.

$f(x,y) = x^2 + y^2$

check $f^*(x,y) = \sqrt{x^2+y^2}$

(if you were asked for value of max/min distance, find pts with x^2+y^2, determine distance with $\sqrt{x^2+y^2}$)

in this question, f^* is not necessary

$$f(x,y) = x^2+y^2 \qquad f(x,y) = \sqrt{x^2+y^2}$$

$$f\left(0, \frac{1}{\sqrt{3}}\right) = 0^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3} \qquad \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$$

$$f\left(0, \frac{-1}{\sqrt{3}}\right) = \frac{1}{3}$$

$$f\left(\frac{1}{\sqrt{2}}, 0\right) = \left(\frac{1}{\sqrt{2}}\right)^2 + 0^2 = \frac{1}{2} \qquad \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$f\left(\frac{-1}{\sqrt{2}}, 0\right) = \frac{1}{2}$$

still: $\frac{1}{\sqrt{2}} > \frac{1}{\sqrt{3}}$

$\frac{1}{2} > \frac{1}{3}$ so $\left(\pm\frac{1}{\sqrt{2}}, 0\right)$ are pts of maximum

$\left(\pm\frac{1}{\sqrt{3}}, 0\right)$ are pts of minimum